# Unsupervised Learning and Clustering

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# Outline

1. Unsupervised learning and clustering
2. Distance and similarity measures
3. K-means clustering
4. Hierarchical clustering
5. Cluster evaluation
6. Cluster interpretation

# Unsupervised learning and clustering

# Supervised vs. unsupervised learning

- Supervised: we observe predictors and a response, $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$, and learn $f$ such that $y \approx f(\mathbf{x})$
- Unsupervised: we observe only $\mathbf{x}^{(i)} \in \mathbb{R}^p$ for $i = 1, \ldots, n$, collected in the $n \times p$ data matrix $\mathbf{X}$
- No response $y$: no “correct answer”, no test error, no cross-validation of accuracy
- The goal is to discover structure: subgroups of observations, relationships among variables, low-dimensional summaries

# Clustering: the goal

- Partition the $n$ observations into $K$ groups $C_1, \dots, C_K$ so that observations within a group are similar and observations in different groups are different
- Hard partition: $C_1 \cup \cdots \cup C_K = \{1, \dots, n\}$ and $C_j \cap C_k = \emptyset$ for $j \neq k$
- Two ingredients: how to measure similarity (a distance) and how to search for a good partition (an algorithm)
- Examples: customer segmentation, grouping cell lines by gene expression, grouping districts by socio-economic indicators

# Distance and similarity measures

# Dissimilarity and similarity

- A dissimilarity $d(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) \geq 0$ is small when two observations are alike; clustering only needs the $n \times n$ matrix of pairwise dissimilarities
- A distance (metric) also satisfies $d(\mathbf{x}, \mathbf{x}') = 0 \iff \mathbf{x} = \mathbf{x}'$, symmetry $d(\mathbf{x}, \mathbf{x}') = d(\mathbf{x}', \mathbf{x})$, and the triangle inequality

$$d(\mathbf{x}, \mathbf{x}'') \leq d(\mathbf{x}, \mathbf{x}') + d(\mathbf{x}', \mathbf{x}'')$$

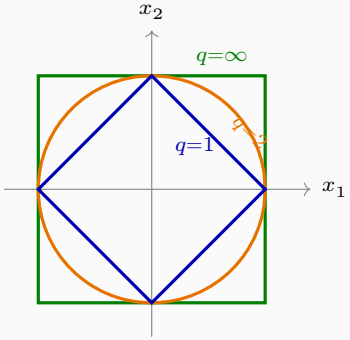
- A similarity $s$ is the opposite (large for alike observations); if $s \in [0, 1]$, then $d = 1 - s$ is a dissimilarity
- The choice of $d$ defines what "similar" means and can change the clusters more than the choice of algorithm

# Minkowski distances

- For $\mathbf{x}^{(i)}, \mathbf{x}^{(l)} \in \mathbb{R}^p$,

$$d_q(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = \left( \sum_{j=1}^{p} |x_j^{(i)} - x_j^{(l)}|^q \right)^{1/q}$$

- $q = 2$ Euclidean: the default; squaring gives large gaps more weight
- $q = 1$ Manhattan: sum of absolute differences; less dominated by one large gap
- $q \to \infty$ Chebyshev: $\max_j |x_j^{(i)} - x_j^{(l)}|$; only the worst coordinate counts




Points at distance 1 from the origin

# Example data: USArrests

- Violent crime in the 50 US states in 1973
- No response and no group labels: a natural setting for unsupervised learning
- Variables are on different scales: three arrest rates per 100,000 residents and one percentage

| Variable | Description | Mean | SD |
|---|---|---|---|
| Murder | murder arrests per 100,000 residents | 7.79 | 4.36 |
| Assault | assault arrests per 100,000 residents | 170.76 | 83.34 |
| UrbanPop | percent of the population living in urban areas | 65.54 | 14.47 |
| Rape | rape arrests per 100,000 residents | 21.23 | 9.37 |

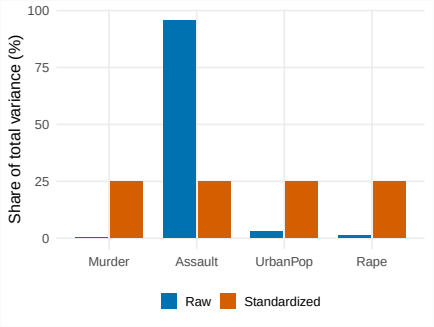
Sources: World Almanac and Book of Facts 1975 (arrests); Statistical Abstracts of the United States 1975 (urban population)

# Scale matters: standardize first

- A variable measured on a large scale dominates the sum of squared (or absolute) differences
- Standardize each variable before computing distances:

$$z_j^{(i)} = \frac{x_j^{(i)} - \bar{x}_j}{s_j}$$

- Exception: variables already on a common, meaningful scale (e.g. counts of the same item)




USArrests: share of total variance

# Mahalanobis distance

- Let $\mathbf{S}$ be the sample covariance matrix of the variables. Then

$$d_M(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = \sqrt{(\mathbf{x}^{(i)} - \mathbf{x}^{(l)})^\top \mathbf{S}^{-1} (\mathbf{x}^{(i)} - \mathbf{x}^{(l)})}$$

- Rescales each variable and accounts for correlation between variables; equals Euclidean distance if $\mathbf{S} = \mathbf{I}$
- Use when variables are strongly correlated, so that Euclidean distance counts the same information several times
- Needs an invertible $\mathbf{S}$ (so $n > p$ and no exact collinearity), and $\mathbf{S}$ is estimated from all data, not within clusters

# Correlation-based distance and cosine similarity

- Correlation between the profiles of observations $i$ and $l$ across the $p$ variables, where $\bar{x}^{(i)}$ is the mean of observation $i$ over its $p$ variables:

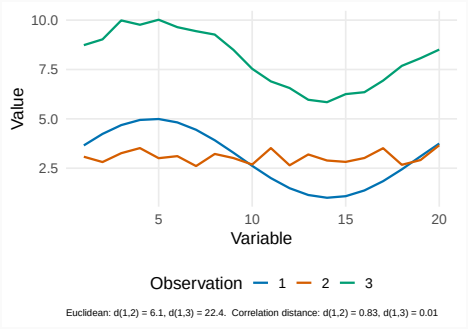
$$r_{il} = \frac{\sum_j (x_j^{(i)} - \bar{x}^{(i)})(x_j^{(l)} - \bar{x}^{(l)})}{\sqrt{\sum_j (x_j^{(i)} - \bar{x}^{(i)})^2}\sqrt{\sum_j (x_j^{(l)} - \bar{x}^{(l)})^2}}, \qquad d_{\text{cor}}(i,l) = 1 - r_{il}$$

- Cosine similarity is the same quantity without centering:

$$\cos\theta_{il} = \frac{\mathbf{x}^{(i)\top}\mathbf{x}^{(l)}}{\|\mathbf{x}^{(i)}\|\,\|\mathbf{x}^{(l)}\|}$$

# When shape matters more than level

- Observations 1 and 3 have the same profile shape at different levels; observation 2 is flat
- Euclidean distance calls 1 and 2 close and 1 and 3 far apart; correlation distance says the opposite
- Use it when profile shape matters, not level (gene expression, shopping baskets, text); variables must be on comparable scales




Simulated: 3 observations on 20 variables. 1 is a sine curve, 3 is observation 1 shifted up by 5, 2 is flat noise around 3.

# Binary and categorical variables

- For two binary vectors, count the variables where they are $(1,1)$, $(1,0)$, $(0,1)$ and $(0,0)$: $n_{11}$, $n_{10}$, $n_{01}$, $n_{00}$, with $p = n_{11} + n_{10} + n_{01} + n_{00}$
- Simple matching: $s = \dfrac{n_{11} + n_{00}}{p}$
  - use when $0$ and $1$ are equally informative (symmetric), e.g. sex, urban/rural
- Jaccard: $s = \dfrac{n_{11}}{n_{11} + n_{10} + n_{01}}$
  - use when only presence is informative, e.g. rare symptoms, species present, items purchased
- Categorical variables with more than two levels: $s$ is the fraction of variables on which the two observations agree; distance is $d = 1 - s$

# Mixed data: Gower dissimilarity

- Survey data typically mix continuous, ordinal and categorical variables; Gower combines per-variable similarities

$$s_{il} = \frac{\sum_{j=1}^{p} w_j^{il}\, s_j^{il}}{\sum_{j=1}^{p} w_j^{il}}, \qquad d_{il} = 1 - s_{il}$$

- Continuous: $s_j^{il} = 1 - |x_j^{(i)} - x_j^{(l)}|/R_j$, where $R_j$ is the range of variable $j$
- Categorical: $s_j^{il} = 1$ if $x_j^{(i)} = x_j^{(l)}$ and $0$ otherwise; ordinal variables can use their ranks
- $w_j^{il} = 0$ if variable $j$ is missing for $i$ or $l$, else $1$

# Choosing a dissimilarity: summary

| Measure | Variables | Use when |
|---|---|---|
| Euclidean | continuous | default; variables standardized, compact groups |
| Manhattan | continuous, counts | robustness to a single large difference |
| Mahalanobis | continuous | strongly correlated variables, $n > p$ |
| Correlation, cosine | continuous | profile shape matters more than level |
| Matching | binary, categorical | $0$ and $1$ equally informative |
| Jaccard | binary | only presence is informative |
| Gower | mixed | continuous, ordinal and categorical variables together |

# K-means clustering

# K-means: the objective

- $K$ is the number of clusters, chosen in advance; each cluster $C_k$ is summarized by its mean (centroid), hence "K-means":

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}^{(i)}$$

- Find the partition that minimizes the total within-cluster sum of squares

$$\min_{C_1,\ldots,C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \left\| \mathbf{x}^{(i)} - \bar{\mathbf{x}}_k \right\|^2$$

- Equivalent (up to a factor of 2) to making pairwise squared distances within each cluster small: $\frac{1}{|C_k|} \sum_{i,l \in C_k} \|\mathbf{x}^{(i)} - \mathbf{x}^{(l)}\|^2$

# K-means: the algorithm

There are almost $K^n/K!$ partitions: exhaustive search is impossible, so we need an algorithm.

1. Randomly assign each observation to one of $K$ clusters
2. Repeat until the assignments stop changing:
   (a) compute the centroid $\bar{\mathbf{x}}_k$ of each cluster
   (b) assign each observation to the cluster with the closest centroid (Euclidean distance)

- This procedure is known as Lloyd's algorithm
- Each pass is cheap, $O(npK)$, so K-means scales to large $n$

# Why the algorithm decreases the objective

- For any cluster $C$ and any point $\mathbf{m}$,

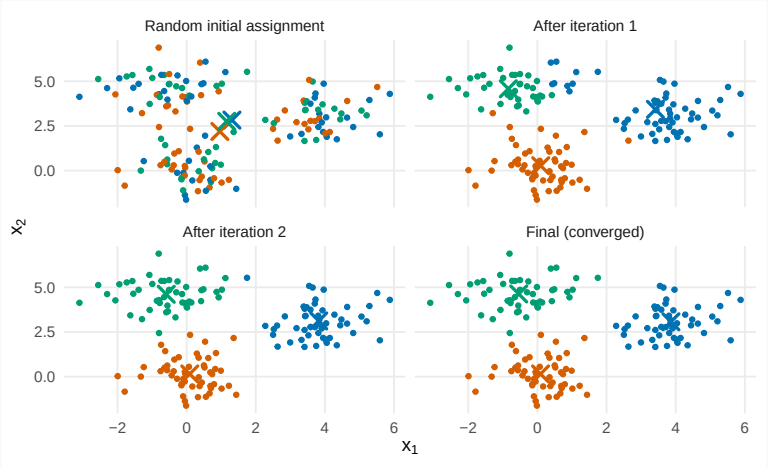
$$\sum_{i \in C} \|\mathbf{x}^{(i)} - \mathbf{m}\|^2 = \sum_{i \in C} \|\mathbf{x}^{(i)} - \bar{\mathbf{x}}_C\|^2 + |C| \, \|\bar{\mathbf{x}}_C - \mathbf{m}\|^2$$

- Step (a): the centroid is the best center for its cluster, so the objective cannot increase
- Step (b): moving an observation to its nearest centroid reduces its squared distance, so the objective cannot increase
- The objective is bounded below and there are finitely many partitions, so the algorithm converges, but only to a local optimum: a partition that neither step can improve, which need not be the best partition overall

# Example data: simulated clusters

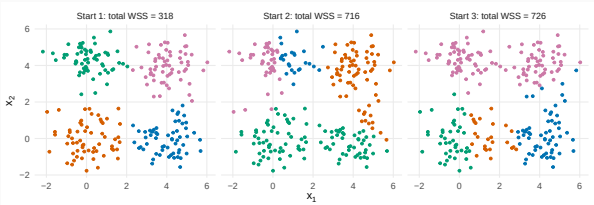
- The K-means examples, and some later ones, use simulated data with two variables, $x_1$ and $x_2$, so the true groups are known and can be plotted
- Each group is a round normal cloud: both coordinates are drawn independently from $N(\mu, \sigma^2)$ around the group's center
- Three-cluster data (K-means iterations): 3 groups of 50 observations, centers $(0, 0)$, $(4, 3)$ and $(-0.5, 4.5)$, $\sigma = 0.9$
- Four-cluster data (multiple starts, and later choosing $K$): 4 groups of 60 observations at the corners of a square, $(0, 0)$, $(4, 0)$, $(0, 4)$ and $(4, 4)$, $\sigma = 0.8$
- Smaller simulated sets used for single examples are described on the slides where they appear

# K-means in action




Three-cluster simulated data, $K = 3$. Crosses mark the centroids.

# Local optima and multiple starts




- Four-cluster data, $K = 4$, three single starts: the solution depends on the random initial assignment, and different starts can end in different local optima
- Run the algorithm from many random starts and keep the solution with the smallest objective
- Smarter initialization, k-means++: choose the initial centroids one at a time, each new one more likely to be a point far from those already chosen

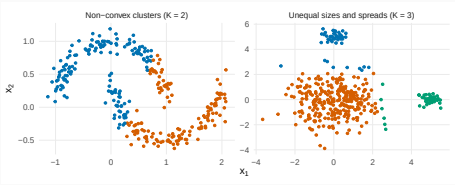
# Choosing $K$ and assigning new observations

- $K$ must be specified before running the algorithm; the evaluation section covers how to choose it
- The objective always decreases as $K$ grows, so it cannot be minimized over $K$
- A new observation $\mathbf{x}^*$ is assigned to the nearest centroid:

$$\hat{k}^* = \arg\min_k \left\| \mathbf{x}^* - \bar{\mathbf{x}}_k \right\|$$

- Standardize $\mathbf{x}^*$ using the means and standard deviations of the data used to fit the clusters

# Strengths and limitations of K-means




Simulated. Left: two interleaved half-moons of 150 points each. Right: one large, spread-out group (300 points, $\sigma = 1.2$) and two small, tight groups (40 points each, $\sigma = 0.3$).

- Strengths: simple, fast, works well for compact, roughly spherical clusters of similar size
- Limitations: non-convex shapes (such as the half-moons), very unequal sizes or spreads, and outliers (centroids are means) can give poor clusters
- Also requires $K$, depends on the starting point, and assigns every observation to a cluster

# Hierarchical clustering

# Hierarchical clustering: the idea

- K-means needs $K$ in advance; hierarchical clustering does not
- It builds a tree of nested clusters, the dendrogram, from which any number of clusters can be read off
- Agglomerative (bottom-up) is the most common: start with $n$ singleton clusters and repeatedly fuse the two closest clusters
- Needs a dissimilarity between observations and a rule for the dissimilarity between clusters (the linkage)

# Agglomerative algorithm

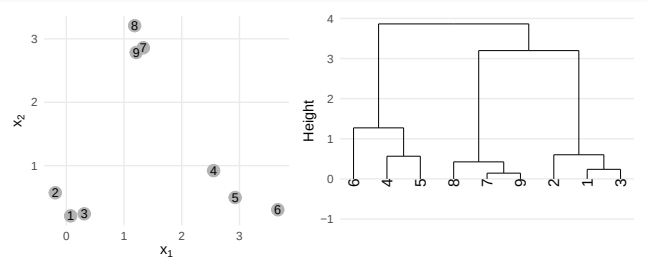
1. Start with $n$ clusters, one per observation, and compute all $\binom{n}{2}$ pairwise dissimilarities
2. For $i = n, n-1, \ldots, 2$:
   a. find the pair among the $i$ clusters with the smallest inter-cluster dissimilarity and fuse them; the dissimilarity is the height of the fusion in the dendrogram
   b. recompute the inter-cluster dissimilarities among the remaining $i-1$ clusters

# Linkage: dissimilarity between clusters

Let $A$ and $B$ be two clusters and $d_{il} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(l)})$.

- Complete: $d(A, B) = \max_{i \in A,\, l \in B} d_{il}$
- Single: $d(A, B) = \min_{i \in A,\, l \in B} d_{il}$
- Average: $d(A, B) = \dfrac{1}{|A|\,|B|} \sum_{i \in A} \sum_{l \in B} d_{il}$
- Centroid: $d(A, B) = d(\bar{\mathbf{x}}_A, \bar{\mathbf{x}}_B)$, the dissimilarity between the two cluster centroids
- Ward: fuse the pair of clusters whose merger increases the total within-cluster sum of squares the least (the K-means objective)

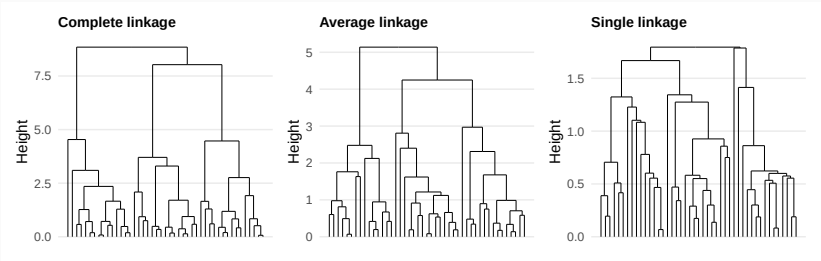
# Reading a dendrogram




Simulated: 9 points in three groups of three; Euclidean distance, complete linkage

- Each leaf is an observation; the height of a fusion is the dissimilarity at which the two clusters joined
- Similarity of two observations is judged by the height where they first fuse, not by how close their leaves are horizontally
- Cutting the tree at a height gives a partition into clusters

# Effect of linkage



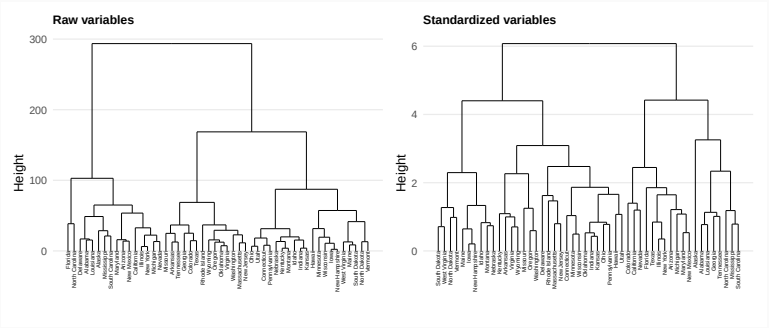

Simulated: 45 points in three groups of 15 ($\sigma = 1$); Euclidean distance

- Complete and average: compact, fairly balanced clusters; complete is the usual default
- Single: chaining, where clusters grow by adding one point at a time; useful for elongated groups, but gives trailing, unbalanced trees
- Centroid: popular in genomics, but can produce inversions (a fusion lower than an earlier one)

# Cutting the tree

- Choose the number of clusters $K$ or a height $h$ at which to cut
- A large gap between successive fusion heights suggests a natural place to cut, but this is a judgment call
- Clusters are nested: those for $K = 2$ are merged clusters of $K = 3$
- Nesting can be unrealistic: a population split by sex and by nationality does not form a single hierarchy

# Effect of scaling on the clusters

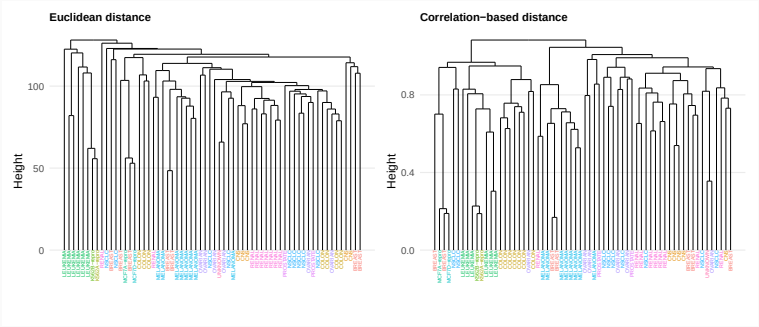



- USArrests, Euclidean distance, complete linkage: on raw data the tree is driven almost entirely by Assault, the variable with by far the largest variance
- After standardizing, all four variables contribute and the grouping of states changes

# Example data: NCI60

- Gene expression for 64 human cancer cell lines (tumor cells grown in the laboratory) from the US National Cancer Institute's NCI-60 panel, measured with microarrays, which record the activity of thousands of genes at once (Ross et al., 2000)
- $p = 6{,}830$ genes per cell line, far more variables than observations ($p \gg n$), so Mahalanobis distance cannot be used; each gene is standardized before computing distances
- Each cell line carries a cancer-type label (14 labels):
  - breast 7, CNS 5, colon 7, leukemia 6, melanoma 8, non-small cell lung 9, ovarian 6, prostate 2, renal 9
  - 4 replicate samples (`K562A-repro`, `K562B-repro`, `MCF7A-repro`, `MCF7D-repro`) and 1 `UNKNOWN`, one cell line each
- The labels are not used to build the clusters; they are used afterwards to check whether the clusters match cancer types

# Example: NCI60 gene expression




- Leaves are labeled and colored by cancer type, which was not used to build the tree
- Average linkage with Euclidean distance (left) and correlation-based distance between cell lines (right) group the cell lines differently

# Cluster evaluation

# Why evaluating clusters is hard

- Without a response there is no ground truth, and every clustering algorithm returns clusters even when the data have no cluster structure.
- Evaluation therefore asks several questions: how many clusters, how compact and well separated they are, whether they agree with known labels, and whether they are stable.
- Internal measures use only the data and the partition; external measures compare with known labels; stability checks whether the result survives small changes.

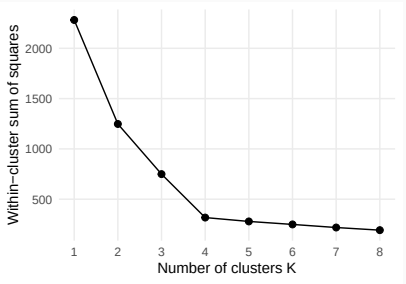
# Internal measure: within-cluster sum of squares

- Total variation: TSS $=$ WSS $+$ BSS ($\bar{\mathbf{x}}$: overall mean)

$$\text{WSS} = \sum_k \sum_{i \in C_k} \|\mathbf{x}^{(i)} - \bar{\mathbf{x}}_k\|^2$$

$$\text{BSS} = \sum_k |C_k| \, \|\bar{\mathbf{x}}_k - \bar{\mathbf{x}}\|^2$$

- BSS/TSS: share of variation explained by the clusters
- WSS always falls as $K$ grows; look for an elbow, a $K$ after which it falls much more slowly




Four-cluster simulated data. The elbow can be ambiguous.

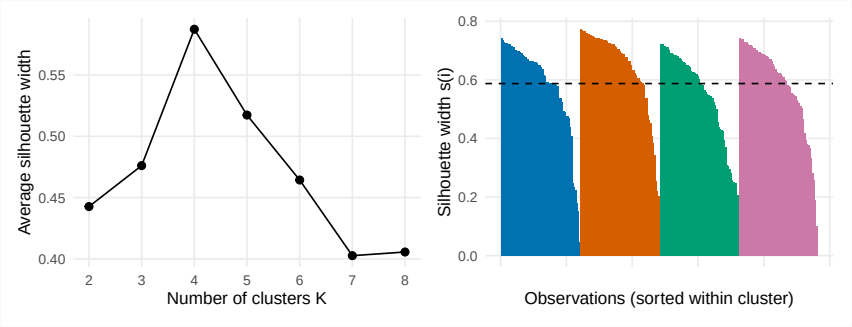
# Internal measure: silhouette

- For observation $i$ in cluster $C_k$, let $a(i)$ be its mean dissimilarity to the other members of $C_k$, and $b(i)$ the smallest mean dissimilarity to the members of any other cluster

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1]$$

- $s(i)$ is the silhouette width of observation $i$; near $1$: well clustered; near $0$: on the border between clusters; negative: probably in the wrong cluster
- The average silhouette width summarizes a partition; choose the $K$ that maximizes it

# Silhouette in practice




- Four-cluster simulated data. Left: average silhouette width against $K$; right: individual silhouette widths for the chosen $K = 4$
- Unlike WSS, the average silhouette does not always improve with $K$, so it can point to a best $K$

# Hierarchical clustering: cophenetic correlation

- The cophenetic distance $t_{il}$ between observations $i$ and $l$ is the height at which they first join in the dendrogram
- The cophenetic correlation is the correlation between the original dissimilarities and the cophenetic distances over all pairs:

$$c = \mathrm{cor}(\{d_{il}\}_{i<l}, \{t_{il}\}_{i<l})$$

- Close to $1$: the dendrogram represents the original dissimilarities faithfully
- Use it to compare linkages for the same dissimilarity

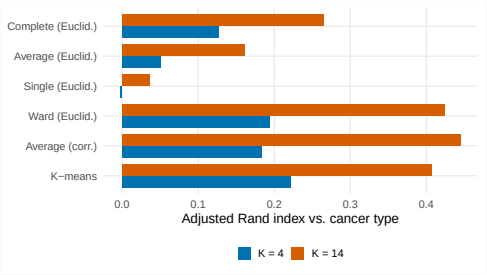
# External measure: Rand index and adjusted Rand index

- When reference labels exist, compare the partition $U$ with the labels $V$ through pairs of observations
- Rand index: the fraction of pairs on which $U$ and $V$ agree, i.e. both place the pair together or both place it apart
- The adjusted Rand index corrects for agreement by chance. With $n_{ab}$ the number of observations in cluster $a$ of $U$ and class $b$ of $V$, and $n_{a\cdot}$, $n_{\cdot b}$ the row and column totals,

$$\mathrm{ARI} = \frac{\sum_{a,b}\binom{n_{ab}}{2} - t}{\frac{1}{2}\left(\sum_a \binom{n_{a\cdot}}{2} + \sum_b \binom{n_{\cdot b}}{2}\right) - t}, \qquad t = \frac{\sum_a \binom{n_{a\cdot}}{2}\sum_b \binom{n_{\cdot b}}{2}}{\binom{n}{2}}$$

- $\mathrm{ARI} = 1$ for identical partitions and about $0$ for random labeling (it can be negative)

# NCI60: agreement with cancer type




- ARI between each clustering and the 14 cancer types, for $K = 4$ and $K = 14$ clusters
- Labels are not necessarily the true clusters: a low ARI may reflect a mismatch between the labels and the gene expression, not a bad clustering

# Stability and robustness

- Clustering can be sensitive: removing a few observations or changing a setting may change the clusters
- Re-cluster on subsamples or bootstrap samples and check whether the same groups reappear
- Vary the choices: $K$, standardization, dissimilarity, linkage, and random starts
- Report conclusions that hold across reasonable choices, and treat the rest as uncertain

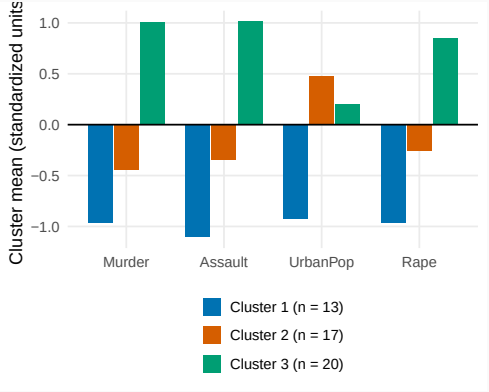
# Evaluation tools: summary

| Tool | Needs labels? | Answers |
|---|---|---|
| WSS and elbow | no | how much variation $K$ clusters explain; choosing $K$ |
| Silhouette | no | how well separated and compact; choosing $K$ |
| Cophenetic correlation | no | how well a dendrogram preserves the dissimilarities |
| Adjusted Rand index | yes | agreement with a reference partition |
| Stability (resampling) | no | whether the clusters are reproducible |

# Cluster interpretation

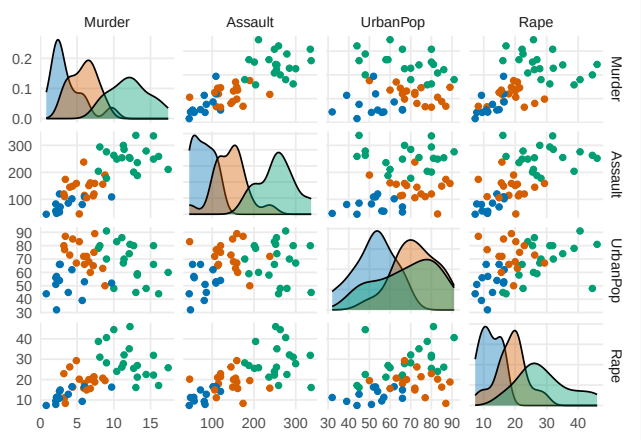
# Profiling the clusters

- Report each cluster's size $|C_k|$ and centroid $\bar{\mathbf{x}}_k$ in original units
- Compare clusters on a common scale with standardized means $(\bar{x}_{kj} - \bar{x}_j)/s_j$; $\bar{x}_{kj}$ is the mean of variable $j$ in cluster $k$
- Variable-by-variable tests (ANOVA) are descriptive only: the clusters were chosen to differ, so $p$-values are not valid




USArrests, K-means with $K = 3$

# Visualizing clusters: pairs plot




- Every pair of the original variables, colored by cluster: shows directly which variables separate which clusters

# Interpreting and reporting clusters

- Name clusters from their profiles, but remember that a cluster is a description of this sample under one set of choices, not a discovered ground truth.
- Report the dissimilarity, standardization, algorithm, $K$, number of starts, and the evidence from evaluation and stability.
- Check the clusters with subject-matter knowledge or new data; do not use the same variables to test whether the clusters differ.
- Cluster labels can be used downstream, for example as a feature in a supervised model.

# Summary

- Clustering partitions unlabeled observations; the result depends on the dissimilarity, scaling, algorithm and $K$
- Standardize unless variables share a meaningful scale; correlation distance for profile shape, Gower for mixed data
- K-means minimizes within-cluster sum of squares with Lloyd iterations, so use many starts; hierarchical clustering gives a dendrogram for any $K$, and the linkage matters
- Evaluate with elbow and silhouette for $K$, cophenetic correlation for dendrograms, ARI when labels exist, and stability always
- Interpret with cluster profiles and visualizations; reading: ISLP Chapter 12, Section 12.4

# Questions?